Data Visualization and Exploratory Data Analysis

Pritha Banerjee

University of Calcutta

banerjee.pritha74@gmail.com

February 24, 2026

Overview

1. Motivations
2. Data Visualization and EDA
   - Data types
3. Non Graphical Analysis
   - Univariate Analysis
   - Multivariate Analysis
4. Graphical Analysis
   - Univariate Analysis
   - Multivariate analysis

Table of Contents

1. Motivations
2. Data Visualization and EDA
   - Data types
3. Non Graphical Analysis
   - Univariate Analysis
   - Multivariate Analysis
4. Graphical Analysis
   - Univariate Analysis
   - Multivariate analysis

AI Vs. ML Vs. Data Science

- **Artificial Intelligence (AI):** make a system/ machine that behave intelligently like humans. Ex: chatbots, voice assistant; AI includes Machine Learning (ML), Robotics, Expert System, Natural Language Processing(NLP), Computer Vision(CV) etc.
- **Machine Learning (ML):** Methods to learn patterns in data (data driven) in order to make AI enabled system. Ex: prediction, recommendation, detection. ML is about Algorithms, models, optimization
- **Data Science:** understanding data to make decisions, generate insights. Ex: Why sales dropped? Tasks are:
  - Cleaning data
  - Exploring patterns
  - Visualization
  - Statistics
  - Sometimes ML

Comparing AI, ML and Data Science

Feature	AI	ML	Data Science
Goal	Make machines intelligent	Enable machines to learn from data	Extract insights from data
Scope	Very broad	Subfield of AI	Interdisciplinary field
Needs data?	Not always	Yes	Yes
Statistics?	Not main focus	Some	Heavy
Linear algebra?	Not main focus	Yes	Yes
Optimization technique?	Not main focus	Yes	Yes
Includes data cleaning?	No	Minimal	Major part
Output	Smart systems	Predictive models	Insights & decisions

Table: AI vs Machine Learning vs Data Science

Role	What You Do	Nature of Work	Focus
AI Engineer	Build intelligent systems (vision, NLP, robotics)	Research + Engineering	Make systems behave intelligently
ML Engineer	Train and deploy ML models	Heavy coding + Math	Build accurate prediction systems
Data Scientist	Analyze data, find insights, build some models	Coding + Analysis + Communication	Understand data and help decisions
Data Analyst	Dashboards, reports, trend analysis	Less coding, more business focus	What happened and why

Table: Career Roles in AI, ML and Data Science

Work Emphasis	Role	Skills
Math + Research + Cutting-edge AI	AI Engineer	ML (strong), Deep Learning, Linear Algebra, Probability, Python, Pytorch,TensorFlow, NLP or CV specialization, Research mindset)
Coding + Systems + Optimization	ML Engineer	Data Structures, Algorithms, Python (advanced),ML algorithms, Mode System design, Deployment (APIs, Docker, cloud), SQL
Analysis + Business + Insights	Data Scientist	Statistics, Data cleaning, Python + Pandas, Visualization tools ML basics, Communication skills
Reports + Dashboards	Data Analyst	Excel / SQL, Basic Python, Dashboards, Business thinking

Table: Primary Work Emphasis, role and skills

Table of Contents

1. Motivations
2. Data Visualization and EDA
   - Data types
3. Non Graphical Analysis
   - Univariate Analysis
   - Multivariate Analysis
4. Graphical Analysis
   - Univariate Analysis
   - Multivariate analysis

Data visualization and Exploratory Data Analysis

- **Data visualization:** The process of representing data graphically using charts, plots, and dashboards. Ex: bar chart, histogram, scatter plot, heat map; visual idea about data.
- **EDA:** The process of analyzing and understanding a dataset before building models through visual methods and summary statistics; emphasizes to understand data; it is required in order to:
  - detect mistakes, check assumptions
  - select appropriate models
  - determine relationships among explanatory variables and outcome variables
- **EDA tasks:** includes Summary statistics (mean, median, std), Checking missing values, Detecting outliers, Finding relationships, Data cleaning, Visualization

Note: Quantitative data can be summarized using summary statistics **non graphically**

Visualization Vs. EDA

Table: Data Visualization vs Exploratory Data Analysis

Aspect	**Data Visualization**	**EDA**
Purpose	To display data visually	To understand data deeply
Scope	Only graphical representation	Broader process
Includes statistics?	Not necessarily	Yes
Includes data cleaning?	No	Yes
Used when?	Anytime (analysis or presentation)	Before modeling
Output	Charts, graphs	Insights + cleaned data

- Drawing only a graph/ plot : visualization
- Interpreting the graph/ plot to reason out : EDA
- EDA uses data visualization

Visualization and types of data

- **Unstructured** raw data from sensors, images, text, voice, video etc. must be processed to a structured form for analysis.
- **Structured data** are typically represented as a table with rows and columns, called rectangular data
- **Structured data** of two types: Numerical (Discrete and Continuous), Categorical (that takes only specific set of values representing categories)
- **Categorical** data could be either Binary (True/ False, 0/1) or Ordinal (explicit ordering like gradation)

Which plot to use?

Carefully look at data type to choose correct plots for visualization

Table: Types of data and appropriate visualizations

Data Type	**Meaning**	**Examples**	**Visualization Type**
Categorical (Nominal)	No order	Gender, Color	Bar chart
Categorical (Ordinal)	Ordered	Grades, Ratings	Ordered bar chart
Numerical (Discrete)	Countable	No. of items	Bar / Histogram
Numerical (Continuous)	Measurable	Height, Age	Histogram, Boxplot

Analysis of data

How many variables you are looking at a time and which plot to use?

- **Univariate Analysis**: One variable
  - **Categorical** variable: typically bar chart to determine which category is most frequent, how is the distribution
  - **Numerical** variable: typically histogram to show distribution and boxplot to show the spread and summary characteristic of data
- **Bivariate Analysis**: Two variables
  - **Both numerical**: scatter plot showing correlation, clusters, outliers; however correlation $\neq$ causation
  - **Numerical and Categorical**: boxplot grouped by category shows median differences, variability; Ex: salary vs. department
  - **Both categorical**: stacked bar chart shows proportion among groups
- **Multivariate Analysis** : Three variables
  - Uses different colors/ sizes to represent third, fourth variables in scatter plots
  - Heatmap to identify strength of correlation between variables, identify redudndant variables

Table of Contents

1. Motivations
2. Data Visualization and EDA
   - Data types
3. Non Graphical Analysis
   - Univariate Analysis
   - Multivariate Analysis
4. Graphical Analysis
   - Univariate Analysis
   - Multivariate analysis

Non graphical univariate analysis (1)

The characteristics of a **population distribution** of a **quantitative** variable is best understood by **center, spread, modality, shape and outliers**

Central tendency:

- **sample mean** of n data samples x_i is $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. It reflects the middle point of the data, the point around which a symmetry holds for the sample distribution; not a good representation if **outlier** exists
- **sample median** of n data samples is an x_i such that 50% of the sample values are less than x_i and 50% are greater; it is robust w. r. t many outliers
- **sample mode** of n data samples is the x_i that occurs the highest number of times; most data is clustered around this value; in distributions can look for bimodality.

Example: 55, 55, 59, 60, 63, 65, 66, 67, 67, 67, 71, 71, 72, 73, 75, 75, 78, 81, 82, 83

Add bias 50 to first 55

See effects on central tendency measures

Non graphical univariate analysis (2)

Spread: spread of univariate sample data

- **sample variance** of n data samples is $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$; different sample variance must match (thus $n-1$)- "unbiased"
- **sample standard deviation** of n data samples is $s = \sqrt{s^2}$; deviations/ spread from the central point (mean)
- **sample range** of n samples is $x_n - x_1$
- Typically, for nice, symmetric distribution the **range** is ± 2 of mean / median.
- **Interquartile Range (IQR):** $Q1, Q2, Q3$ mark the 25%, 50% and 75% points of the data arranged in increasing order; thus Q2 is median, IQR is $Q3 - Q1$; most robus measure of spread

Example: 55, 55, 59, 60, 63, 65, 66, 67, 67, 67, 71, 71, 72, 73, 75, 75, 78, 81, 82, 83

compute measure of spreads.

Non graphical univariate analysis (3)

Skewness and Kurtosis:

- **Skewness:** measure of **asymmetry**; sample skewness is $\gamma_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^{n} (\frac{x_i - \bar{x}}{s})^3$;
- Skewness Interpretation: 0 : symmetric; > 0 : right skewed; < 0 : left skewed
- **Kurtosis:** measure of **peakedness** relative to Gaussian shape; $\gamma_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^{n} \left(\frac{x_i - \bar{x}}{s}\right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}$
- Kurtosis Interpretation: 0 : Normal shape; > 0 : heavy tails, sharp peak (more outliers); < 0 : light tails , flat top

Bivariate: Numerical Vs. Numerical

- For two quantitative variables, use sample **covariance** and /or sample **correlation**
- **covariance:** how much and in which direction one variable changes when the other changes; $Cov(X, Y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n-1}$; Note $Cov(X, X) = Var(x)$
- **Positive covariance** implies if one variable value is above mean, the other will also be above mean, **negative** implies if one variable value is above mean the other is below mean; **value near or equal to** 0 implies two variables vary independently; it is hard to interpret at times. Thus,
- **correlation:** $Cor(X, Y) = \frac{Cov(X,Y)}{s_x s_y}$; where s_x, s_y are standard deviation of X and Y respectively.
- correlation r value being $+1$ implies perfect positive correlation and -1 implies perfect negative correlation, 0 represnting X and Y uncorrelated.
- For **multivariate data**, compute pairwise covariance or correlation in a matrix form.

Example: Covariance and Correlation

Subject ID	Age	Strength	Age-50	Str-19	product
GW	38	20	-12	+1	-12
JA	62	15	+12	-4	-48
TJ	22	30	-28	+11	-308
JMA	38	21	-12	+2	-24
JMO	45	18	-5	-1	+5
JQA	69	12	+19	-7	-133
AJ	75	14	+25	-5	-125
MVB	38	28	-12	+9	-108
WHH	80	9	+30	-10	-300
JT	32	22	-18	+3	-54
JKP	51	20	+1	+1	+1
Total			0	0	-1106

Table 4.4: Covariance Calculation

	X	Y	Z
X	5.00	1.77	-2.24
Y	1.77	7.0	3.17
Z	-2.24	3.17	4.0

Table 4.5: A Covariance Matrix

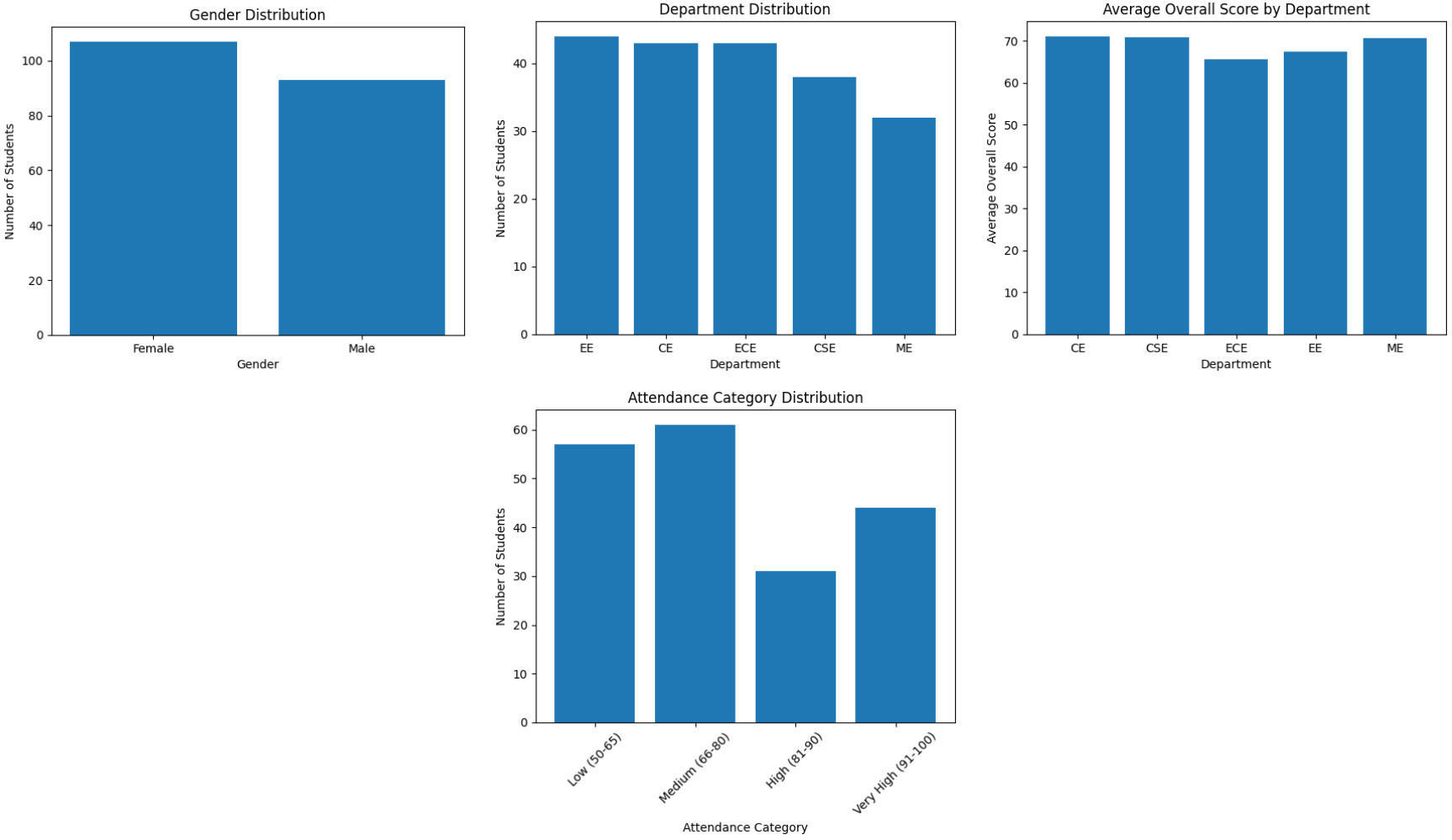
Table of Contents

1. Motivations
2. Data Visualization and EDA
   - Data types
3. Non Graphical Analysis
   - Univariate Analysis
   - Multivariate Analysis
4. Graphical Analysis
   - Univariate Analysis
   - Multivariate analysis

Graphical univariate analysis: Bar chart

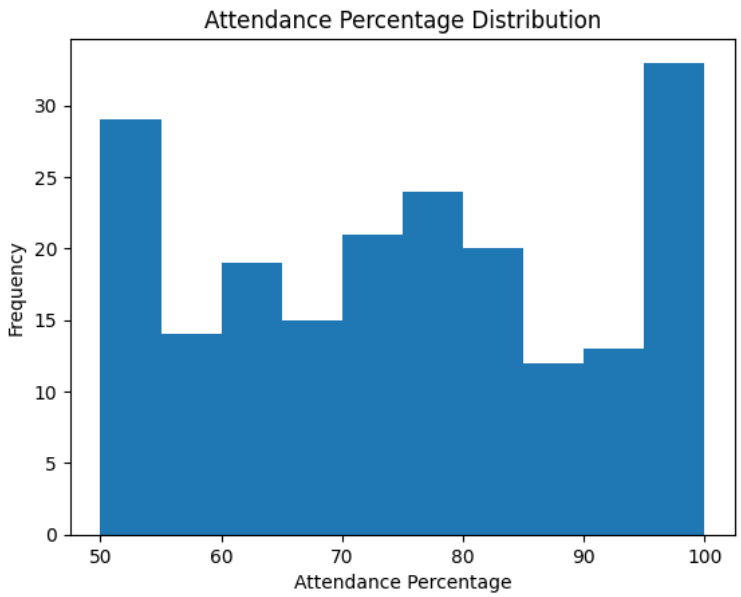
Student ID,Age,Gender,Department,Study Hours per Week,Attendance Percentage, Internal Assessment,Final Exam Score, Overall Score.

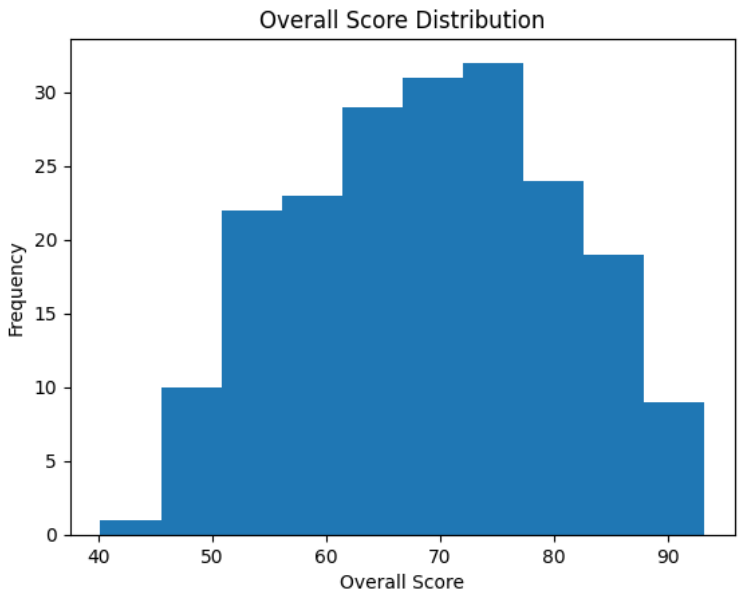
Checking data balance, compare group performance, attendance trends etc.



Graphical univariate analysis: Histogram

- Divide the range of values in sample set into small intervals and count number of observations fall in each interval
- For each interval plot a rectangle with width equal to interval size and height equal to number of observations in that interval
- Detect skewness or clustering, Compare exam performance spread and shape, outliers

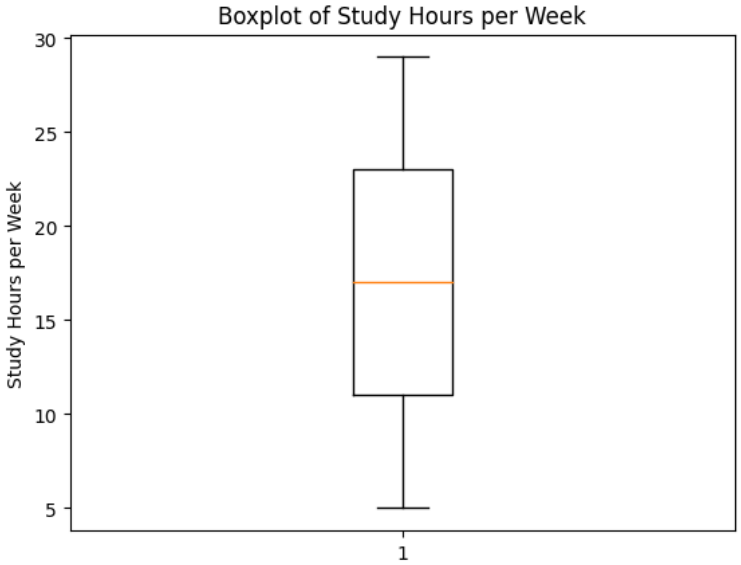


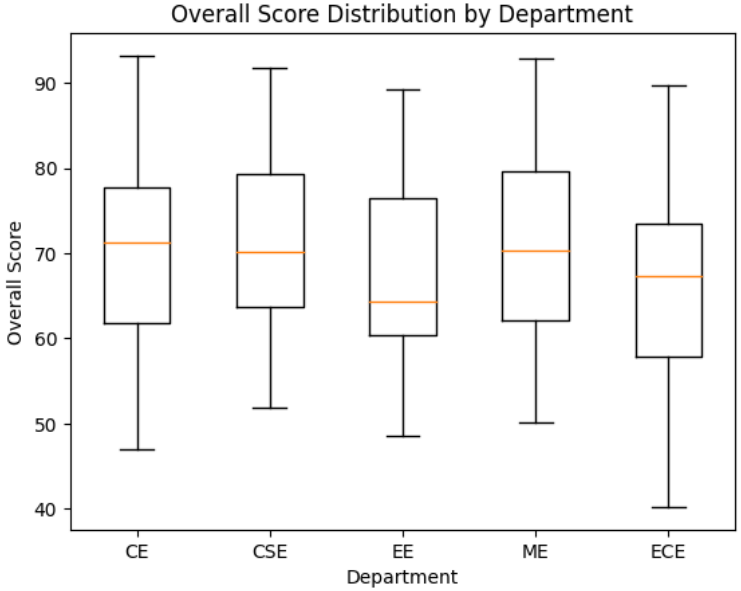


Observations: a) Attendence : skewed at 50, 100 b) Exam score : Normal distribution

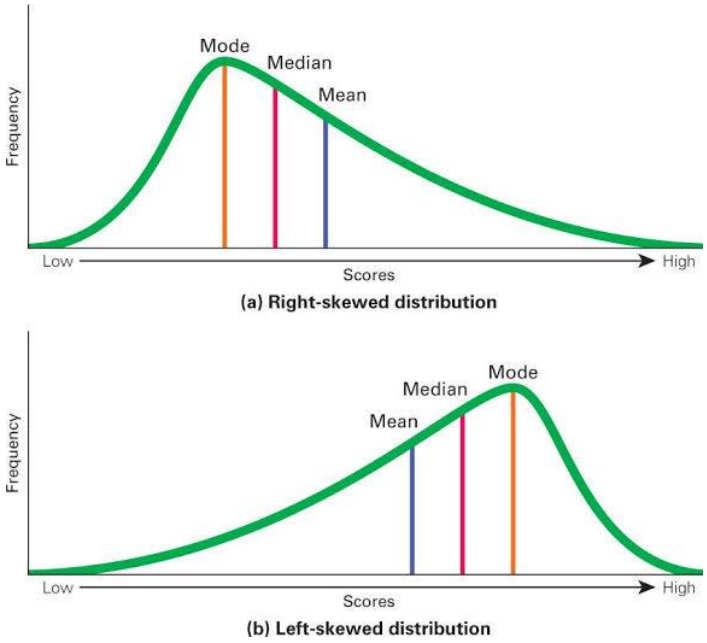
Graphical univariate analysis: Boxplot

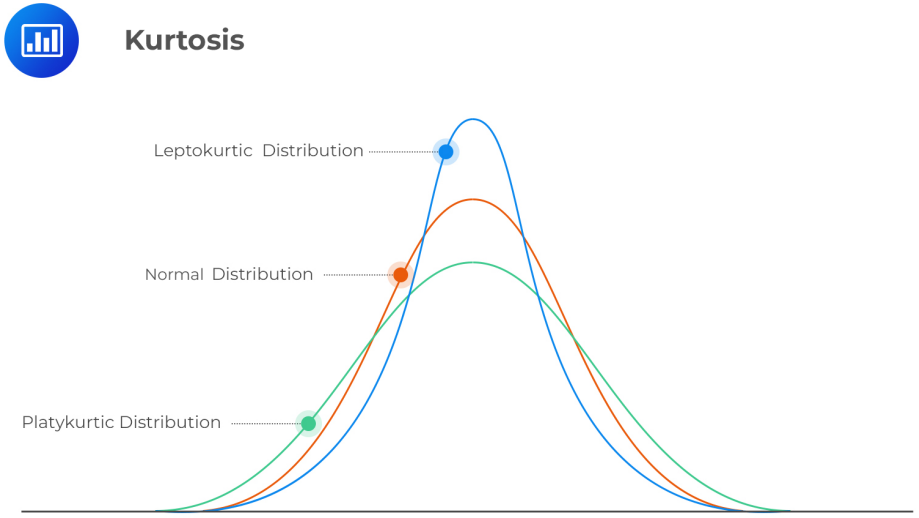
- Find quartiles Q1, Q2 (median), Q3, minimum, maximum, outliers; drawn with whiskers to minimum and maximum
- see Median line, Interquartile range (IQR), Whisker length (spread), Any outliers, Which group performs better (for grouped plots)
- Identify Outliers: compute $IQR = Q3 - Q1$; An outlier is any data point that lies outside the bounds defined by: Lower Bound: $Q1 - 1.5 * IQR$ and Upper Bound: $Q3 + 1.5 * IQR$





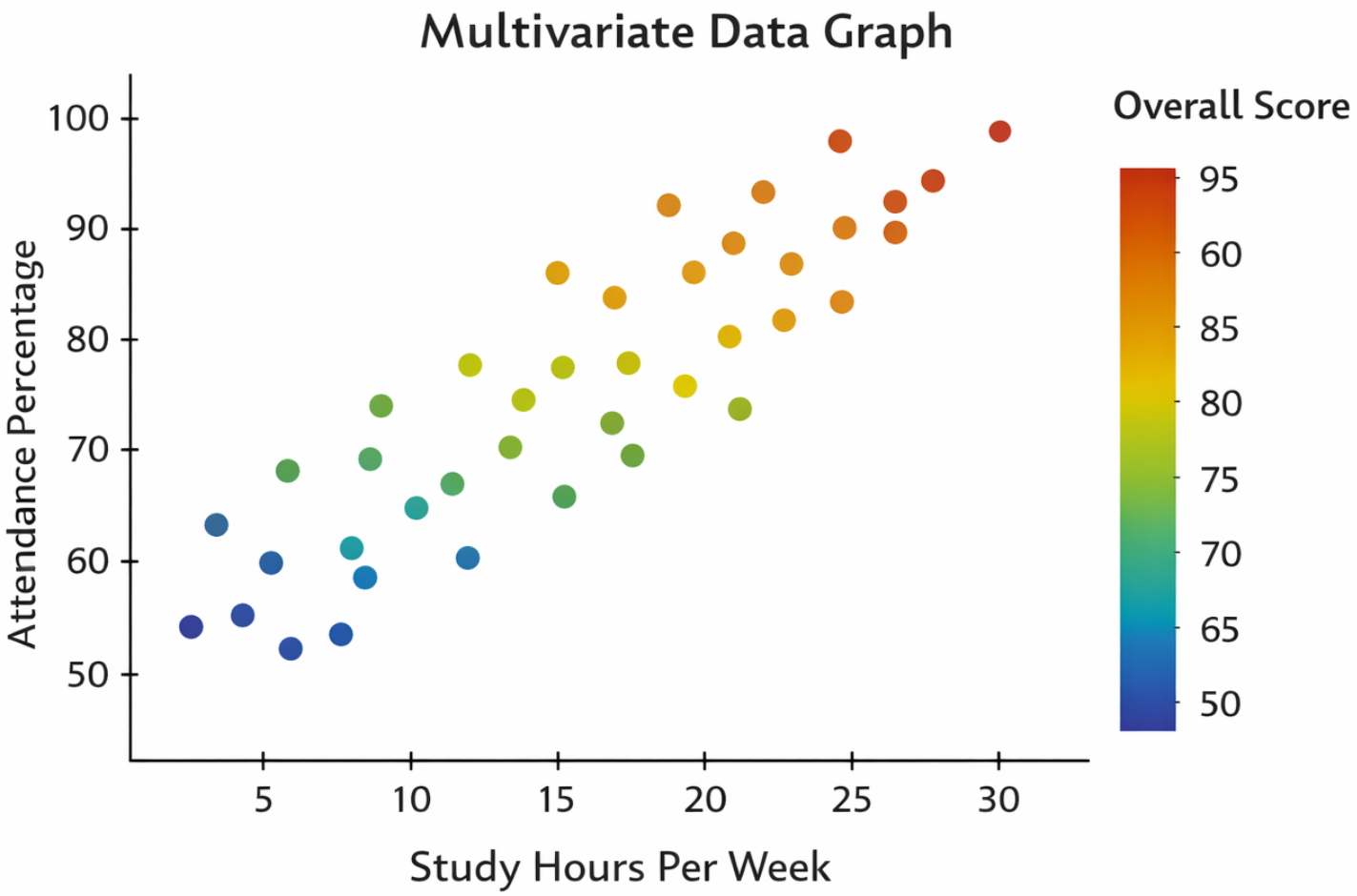
Skewness and Kurtosis:





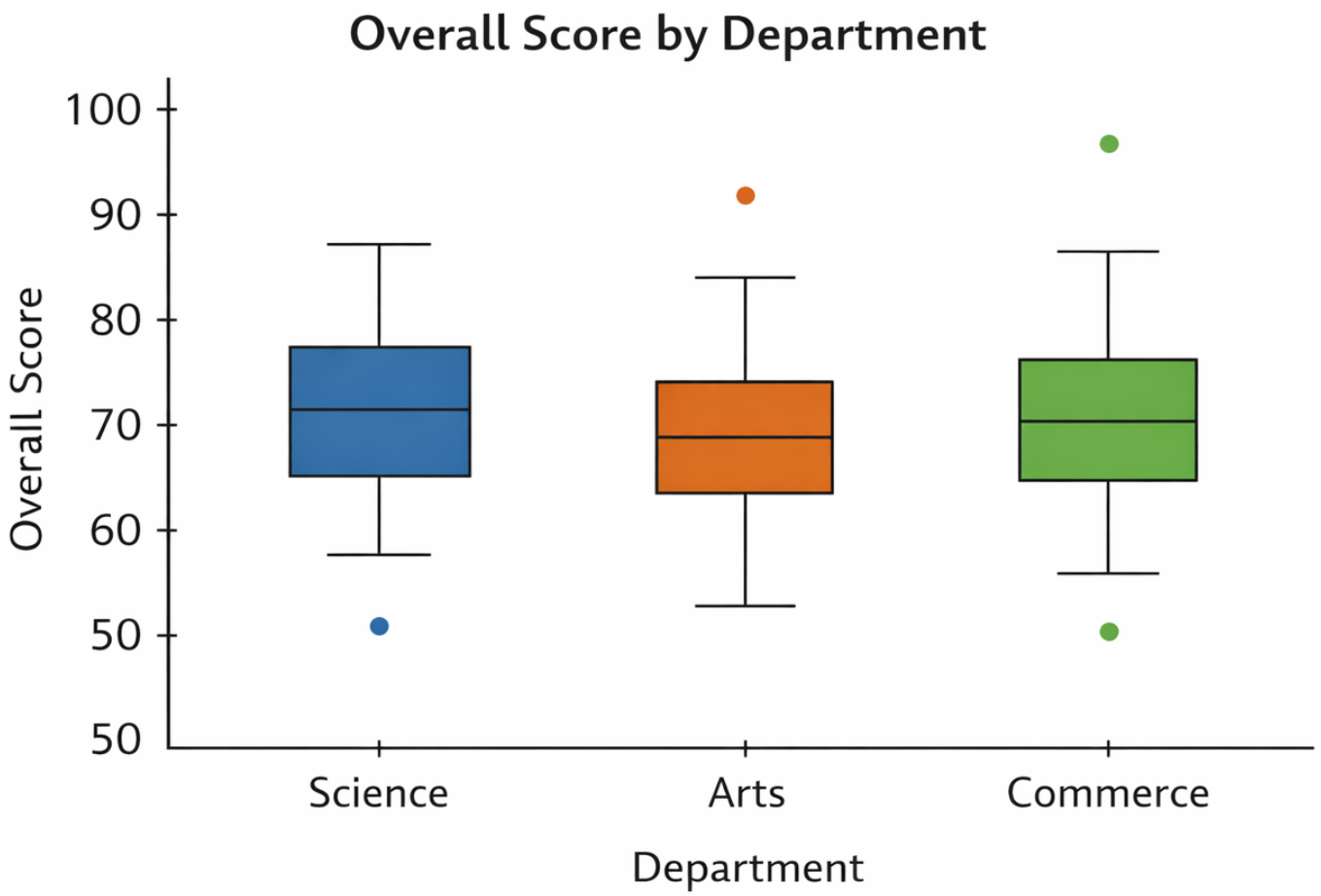
Multivariate : Numerical Vs. Numerical

Scatterplot: shows correlation, clusters and outliers; correlation *ne* causation



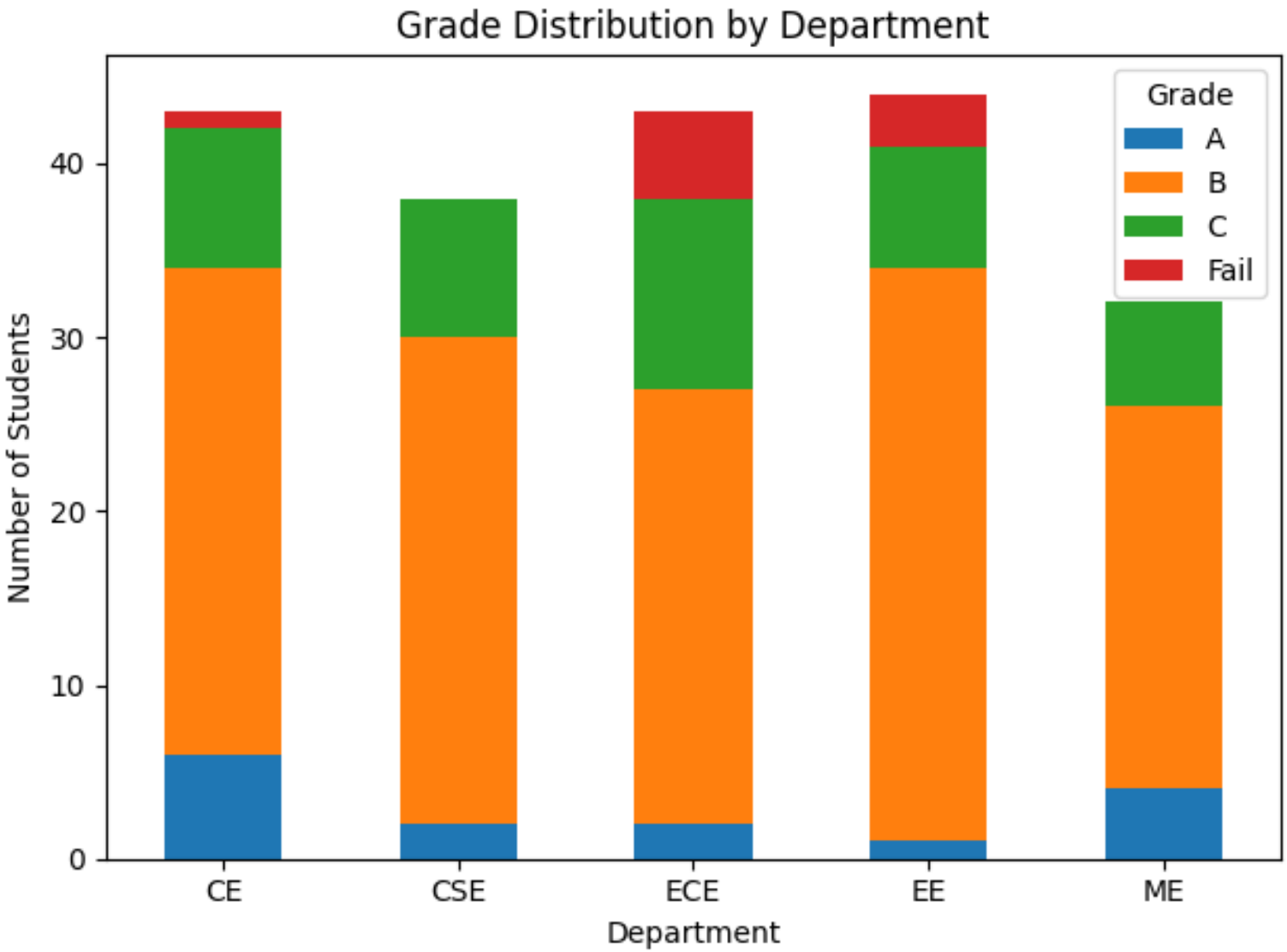
Multivariate: Numerical Vs. Categorical

Boxplot grouped by category: shows median difference and variability



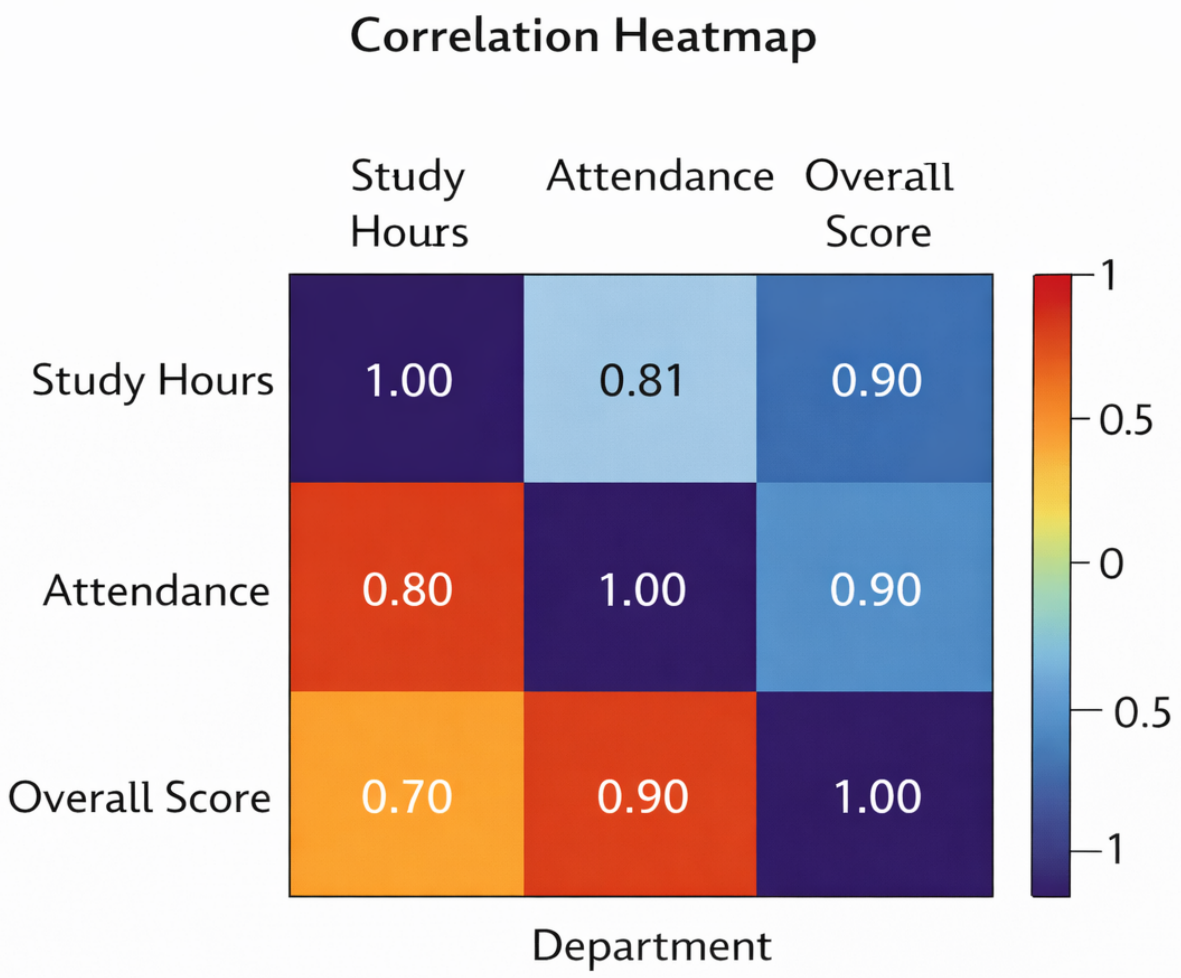
Multivariate: Categorical Vs. Categorical

Stacked bar chart: shows proportions across groups



Multivariate: Correlation Heatmap

shows strength of relationship, thus identifying redundant variables



EDA by visualization

Check quality of data through visualization.

Issue	**How Visualization Helps**
Missing values	Bar plot of null counts
Outliers	Boxplot
Imbalance	Bar chart
Skewness	Histogram
Data entry errors	Strange spikes in plots

Table: Using Visualization to Detect Data Issues

Should give clear titles, proper labels, consistent scales for data visualization.